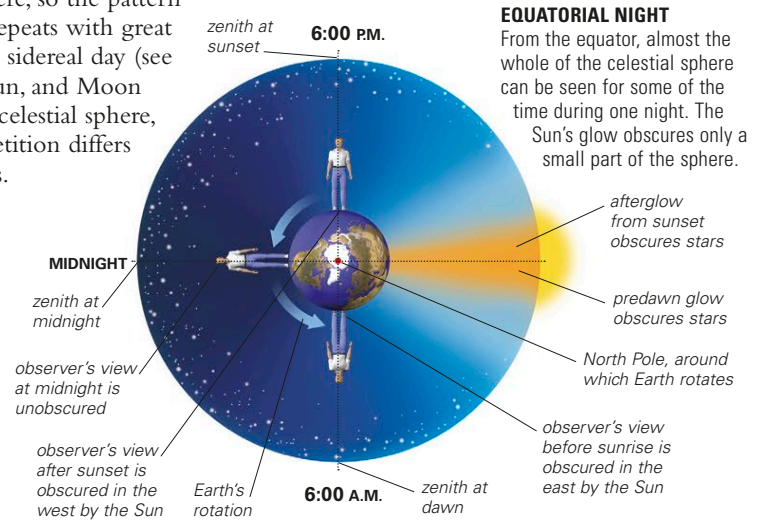




DAILY SKY MOVEMENTS

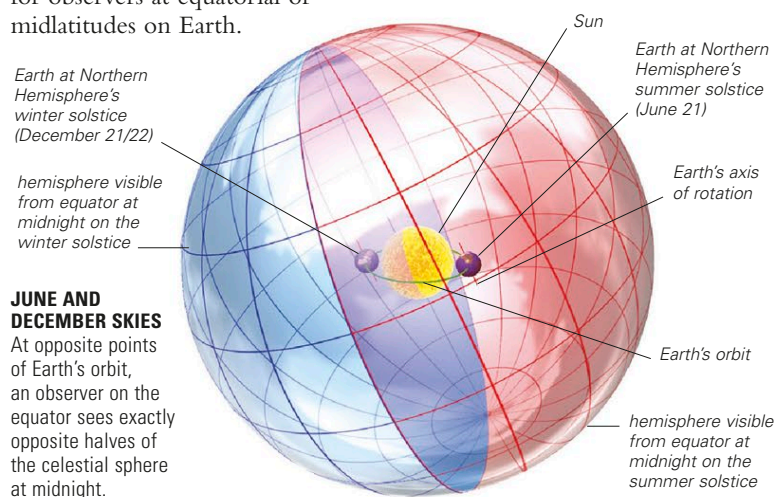
As the Earth spins, all celestial objects move across the sky, although the movements of the stars and planets become visible only at night. For an observer in midlatitudes, stars in polar regions of the celestial sphere describe a daily circle around the north or south celestial pole. The Sun, Moon, planets, and the remaining stars rise along the eastern horizon, sweep in an arc across the sky, and set in the west. This motion has a tilt to the south (for observers in the Northern Hemisphere) or to the north (Southern Hemisphere)—the lower the observer's latitude, the steeper the tilt. Stars have fixed positions on the sphere, so the pattern of their movement repeats with great precision once every sidereal day (see p.66). The planets, Sun, and Moon always move on the celestial sphere, so the period of repetition differs from that of the stars.

CIRCUMPOLAR STARS
Stars in the polar regions of the celestial sphere describe perfect part-circles around the north or south celestial pole during one night, as shown by this long-exposure photograph.



YEARLY SKY MOVEMENTS

As Earth orbits the Sun, the Sun seems to move against the background of stars. As the Sun moves into a region of the sky, its glare washes out fainter light from that part, so any star or other object there temporarily becomes difficult to view from anywhere on Earth. Earth's orbit also means that the part of the celestial sphere on the opposite side to Earth from the Sun—that is, the part visible in the middle of the night—changes. The visible part of the sky at, for example, midnight in June, September, December, and March is significantly different—at least for observers at equatorial or midlatitudes on Earth.

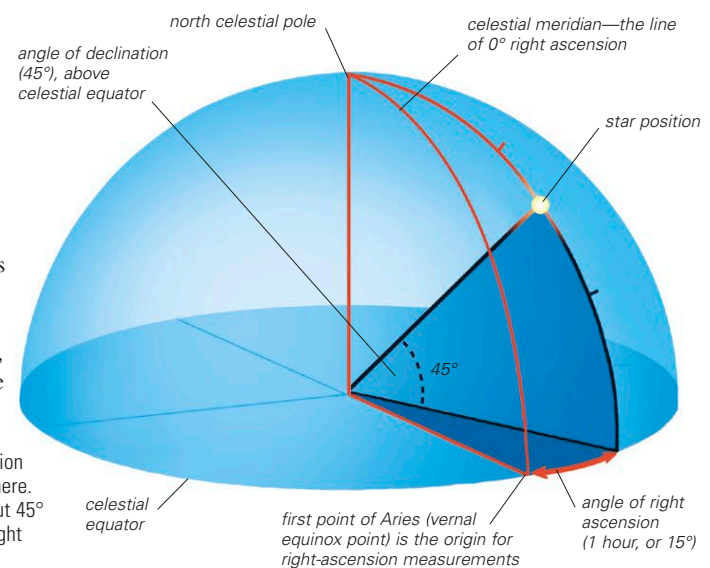


CELESTIAL COORDINATES

Using the celestial sphere concept, astronomers can record and find the positions of stars and other celestial objects. To define an object's position, astronomers use a system of coordinates, similar to latitude and longitude on Earth. The coordinates are called declination and right ascension. Declination is measured in degrees and arc-minutes (60 arc-minutes = 1 degree/1°) north or south of the celestial equator, so it is equivalent to latitude. Right ascension, the equivalent of longitude, is the angle of an object to the east of the celestial meridian. The meridian is a line passing through both celestial poles and a point on the celestial equator called the first point of Aries or vernal equinox point (see p.65). An object's right ascension can be stated in degrees and arc-minutes or in hours and minutes. One hour is equivalent to 15°, because 24 hours make a whole circle.

RECORDING A STAR'S POSITION

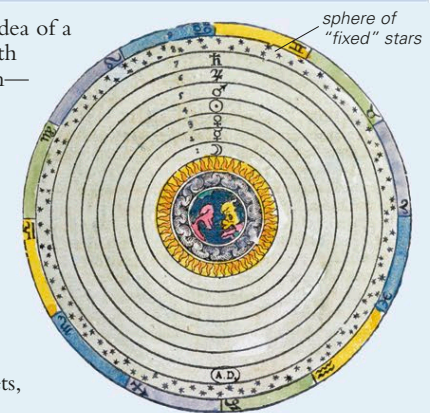
The measurement of a star's position on the celestial sphere is shown here. This star has a declination of about 45° (sometimes written +45°) and a right ascension of about 1 hour, or 15°.



EXPLORING SPACE

ARISTOTLE'S SPHERES

Until the 17th century CE, the idea of a celestial sphere surrounding Earth was not just a convenient fiction—many people believed it had a physical reality. Such beliefs date back to a model of the universe developed by the Greek philosopher Aristotle (384–322 BCE) and elaborated by the astronomer Ptolemy (85–165 CE). Aristotle placed Earth stationary at the universe's center, surrounded by several transparent, concentric spheres to which the stars, planets, Sun, and Moon were attached. Ptolemy supposed that the spheres rotated at different speeds around Earth, producing the observed motions of the celestial bodies.



ARISTOTELIAN MODEL OF THE UNIVERSE
Stars are fixed to the outer sphere. Working inward, the other spheres around Earth carry Saturn, Jupiter, Mars, the Sun, Venus, Mercury, and the Moon.